

9.8 RCC registers

9.8.1 RCC clock control register (RCC_CR)

Address offset: 0x000

Reset value: 0x0000 0063

HSEBYP is cleared upon power-on reset. It is not affected upon other types of reset.

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRIV	Res.	PLLSAI2RDY	PLLSAI2ON	PLLSAI1RDY	PLLSAI1ON	PLLRDY	PLLON	Res.	Res.	Res.	Res.	CSSON	HSEBYP	HSERDY	HSEON
rw		r	rw	r	rw	r	rw					rs	rw	r	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	HSIAFS	HSIRDY	HSIKERON	HSION	MSIRANGE[3:0]				MSIRGSEL	MSIPPLEN	MSIRDY	MSION
				rw	r	rw	rw	rw	rw	rw	rw	rs	rw	r	rw

Bit 31 **PRIV**: RCC privilege

Set and reset by software. This bit can be read by both privileged or unprivileged access. when set, it can only be cleared by a privileged access.

0: RCC registers can be accessed by a privileged or non-privileged access.

1: RCC registers can be accessed only by a privileged access except RCC_AHBxSECSR, RCC_APBx_SECSR and RCC_SECSR.

An unprivileged access to RCC registers is RAZ/WI.

If TrustZone security is enabled (TZEN = 1), when the RCC is not secure, the PRIV bit can be written by a secure or non-secure privileged access.

If the RCC is secure, the PRIV bit can be written only by a secure privileged access. A non-secure write access is ignored.

A secure unprivileged write access on PRIV bit is ignored.

Bit 30 Reserved, must be kept at reset value.

Bit 29 **PLLSAI2RDY**: SAI2 PLL clock ready flag

Set by hardware to indicate that the PLLSAI2 is locked.

0: PLLSAI2 unlocked

1: PLLSAI2 locked

Bit 28 **PLLSAI2ON**: SAI2 PLL enable

Set and cleared by software to enable PLLSAI2.

Cleared by hardware when entering Stop, Standby or Shutdown mode.

0: PLLSAI2 OFF

1: PLLSAI2 ON

Bit 27 **PLLSAI1RDY**: SAI1 PLL clock ready flag

Set by hardware to indicate that the PLLSAI1 is locked.

0: PLLSAI1 unlocked

1: PLLSAI1 locked